

M_1 and M_2

H₀: Data is coming from M_1 . $\left\{ \begin{matrix} f(x, \theta_1) & \& \pi(\theta_1) \end{matrix} \right\}$
 $\uparrow$
 prior

M_1 : ————— M_2 . $\left\{ \begin{matrix} f(x, \theta_2) & \& \pi(\theta_2) \end{matrix} \right\}$

Proper priors for θ_1, θ_2 marginal likelihood M_j is $\int f(\theta_1, \theta_2 | y_j) \pi(\theta_1, \theta_2) d\theta_1 d\theta_2$

$P(M_j | \text{data})$ BF $\rightarrow$ calculation evidence of the compatibility of the data for both the models.

$$B_F = \frac{\frac{\pi(m_1|y)}{\pi(m_1)}}{\frac{\pi(m_2|y)}{\pi(m_2)}} = \frac{m(y|m_1)}{m(y|m_2)}$$

Example: $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$ density function is $f(\theta | \mathbf{X})$

under H_0 : the distribution of $\hat{\theta}$ is degenerate at θ_0 (fixed hypothesis is simple)

under H_1 : $\hat{\theta} \sim q(\theta)$ (prior of θ under H_1)

we need to find the marginal likelihood under both H_0 and H_1 .

Let the parameter space for θ be continuous (i.e. $\mathbb{R}^M$)

Marginal
 likelihood
 under H_0

$$m_0(x) = \int f(x|\theta) p_0(\theta) d\theta$$

And $P(H_0) = P_0$ & $\pi(H_0) = 1 - P_0$

$$\pi(\theta) = P_0 p_0(\theta) + (1 - P_0) p_1(\theta)$$

$$= \frac{P_0 f(x|\theta) \int f(x|\theta) p_0(\theta) d\theta}{\int f(x|\theta) (P_0 p_0(\theta) + (1 - P_0) p_1(\theta)) d\theta}$$

marginal den. for H_0

$$= \left[1 + \left(\frac{1-p_0}{r_0} \right) \frac{n_1(x)}{f(x|p_0)} \right]^{-1} = \left[1 + \frac{1-p_0}{r_0} \frac{1}{\hat{p}_{01}(x)} \right]^{-1}$$

Example: Let $X \sim \text{Binomial}(n, p)$.
and consider testing $H_0: p = p_0$ vs $H_1: p \neq p_0$, $g(x) = 1$
 $n = 5, \alpha = 0.1$, what will be the Bayes factor.
under H_0 : X is binomial $\binom{5}{x} \left(\frac{1}{2}\right)^5$. ~~So the~~
(marginal) $P(X) = \frac{1}{32}$.
 $P(H_0) = P(H_1) = 1$.

Ex: $H_0: p < 0.5$ vs $H_1: p \geq 0.5$
 $X|p \sim \text{Binomial}(n, p)$
 $\pi_0(b) = 2, 0 < p < 0.5$
 $\pi_1(b) = 2, 0.5 < p < 1$
 $n = 100$
 $X = 7$
 $2 \binom{n}{x} p^x (1-p)^{n-x} : H_0$
 $\int_0^1 p^x (1-p)^{n-x} dp : H_1 //$

$$E_n: H_0: X_i \sim N(0,1)$$

$$H_1: X_i \sim \text{Cauchy}(0,1) \quad (f(x|0) = \frac{1}{\pi(1+x^2)})$$

$$(\alpha = (0.2, 0.4, 0.7, 0.83)) \quad (\beta_{F,0} = 0.43)$$

$$\frac{\text{Posterior odds}}{\text{Prior odds}} = \text{Bayes factor} \quad P(H_0) \& P(H_1)$$
$$\text{Ex. n.} \quad \begin{aligned} H_0: X_i &\sim N(\mu, \sigma^2) & \pi_1(\mu, \sigma^2) \\ H_1: X_i &\sim \text{unif}(\mu, \sigma) & \pi_2(\mu, \sigma^2) \end{aligned} \quad \text{Bayes: } \frac{\int L_1(\mu, \sigma) \pi_1(\mu, \sigma) d\mu d\sigma}{\int L_2(\mu, \sigma) \pi_2(\mu, \sigma) d\mu d\sigma}$$

Ex: $X_1, \dots, X_n | \mu \sim N(\mu, 1)$ | $H_0: \begin{cases} X_i | \mu \sim N(\mu, 1) \\ \mu \sim N(0, 1) \end{cases} \hookrightarrow$ Conjugate bay

$$H_0: \mu \sim N(0, 100) \\ n=10, \mu=0.5 \\ BF_{10} = \frac{\phi(\bar{x}; 0, 100)}{\phi(\bar{x}; 0, 1)} \\ \text{Occam's Razor} \quad \text{You pay a price for the flexibility of the model}$$

$$\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$$

$$H_1: \begin{cases} x_i | \mu \sim N(\mu, 1) \\ \mu \sim N(0, 100) \end{cases} \text{Bayes factor} \\ \bar{x} | \mu \sim N(\mu, \frac{1}{n}), \mu \sim N(0, 100) \\ \Rightarrow \bar{x} | H_1 \sim N(0, \frac{1}{n} + \frac{1}{100}) \\ \therefore \bar{x} | H_1 \sim N(0, 100 + \frac{1}{n}) \\ \bar{x} | H_0 \sim N(0, 100 + \frac{1}{n})$$

Bayes Factor consistency: Can we select the correct hypothesis.

$$\text{Ex: } x_1, \dots, x_n | \mu \sim N(\mu, 1)$$

using Bayes Factor?

$$H_0: \mu = 0 \text{ vs } H_1: \mu \neq 0 \\ \mu \sim N(0, \tau^2)$$

Consistency: if $n \rightarrow \infty$, your inference should be correct

$$\text{if } x_1, \dots, x_n \sim N(0, 1) \text{ then } BF_{10,n} \rightarrow \begin{cases} 0 \\ \infty \end{cases} \text{ a.s.}$$

$$\begin{aligned} \mu=0 &\Rightarrow BF_{10,n} \rightarrow 0 \text{ a.s.} \\ \mu \neq 0 &\Rightarrow BF_{10,n} \rightarrow \infty \text{ a.s.} \end{aligned} \quad (BF_{10,n}) \cdot f(x_1, \dots, x_n)$$

$$\begin{aligned} \mu \leq 0 &\quad \mu > 0 \\ N(0, \tau^2) &\quad N(0, \tau^2) \\ 1/0, \infty &\quad 1/0, \infty \end{aligned}$$

$$\bar{x} | H_0 \sim N(0, \frac{1}{n}), \quad \bar{x} | H_1 \sim N(0, \tau^2 + \frac{1}{n})$$

$$BF_{10,n} = \frac{\phi(\bar{x}; 0, \tau^2 + \frac{1}{n})}{\phi(\bar{x}; 0, \frac{1}{n})} \Rightarrow \log BF_{10,n} = \left(-\frac{1}{2} \log(1 + n\tau^2) \right) + \frac{\bar{x}^2 n^2 \tau^2}{2(n\tau^2 + 1)}$$

$$x_1, \dots, x_n \sim N(0, 1) \Rightarrow \sqrt{n} \bar{x} \sim N(0, 1) \\ \Rightarrow n \bar{x}^2 \sim \chi^2_{(1)}$$

$$\therefore \mu=0 \Rightarrow BF_{10,n} \rightarrow 0 \text{ a.s.}$$

$$\Rightarrow P(H_0 | x_1, \dots, x_n) \rightarrow 1 \text{ if } \mu \neq 0 \text{ in the true data-generating process. a.s.}$$

$$\log BF_{10,n} = -\frac{1}{2} \log(1 + n\tau^2) + \frac{\bar{x}^2 n^2 \tau^2}{2(n\tau^2 + 1)}$$

$$\text{Let } x_1, \dots, x_n \sim N(\mu^*, 1) \text{ \& } \mu^* \neq 0 \quad \bar{x} \rightarrow \mu^* \Rightarrow \bar{x}^2 \rightarrow (\mu^*)^2$$

$$\therefore \log BF_{10,n} \xrightarrow{p} \infty \quad n \bar{x}^2 \rightarrow \infty$$

$$n \bar{x}^2 \rightarrow \chi^2_{(1)}$$

$$\therefore P(H_1 | X_1, \dots, X_n) \xrightarrow{p} 1.$$

$$\begin{matrix} H_0: \mu = 0 & \text{vs} & H_1: \mu = 1 \\ \text{But } \mu(2,1) \end{matrix}$$

Ex: ~~$X_1, \dots, X_n \sim N(\mu, 1)$~~

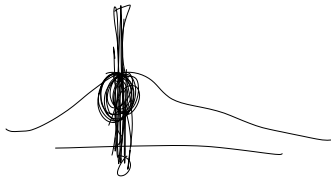
$$H_0: \mu \sim N(9, 1) \quad H_1: \mu \sim N(-9, 1)$$

$$P(H_0 | \text{Data}) + P(H_1 | \text{Data}) = 1$$

Let the data-generating process be ~~$X_i \sim N(10, 1)$~~

$$BF_{01,n} = \frac{\phi(\bar{X}_n; 9, 1 + \frac{1}{n})}{\phi(\bar{X}_n; -9, 1 + \frac{1}{n})} = \exp\left\{ \frac{18\bar{X}_n}{1 + \frac{1}{n}} \right\} \rightarrow \exp\{180\}$$

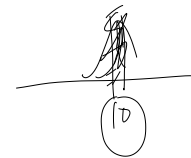
$$\therefore P(H_0 | X_1, \dots, X_n) < 1 \text{ as } n \rightarrow \infty$$



Ex: ~~$H_0: \mu \leq 0$~~ $H_0: \mu \leq 0 \quad H_1: \mu > 0$

$$\begin{matrix} \mu \sim TN(0, 1) & \mu \sim TN(0, 1) \\ (-\infty, 0] & (0, \infty) \end{matrix}$$

Data generating process: $X_1, \dots, X_n \sim N(10, 1)$



Solve it

Ex: $H_0: \mu \leq 0 \quad H_1: \mu > 0$

$$\begin{matrix} \mu \sim TN(0, 1) & \mu \sim TN(0, 1) \\ (-\infty, 0] & (0, \infty) \end{matrix}$$

$$X_1, \dots, X_n \sim N(0, 1)$$

Simulation (3 marks)
Write the details of simulation

Q2 (2 marks)

a) Show that as $n \rightarrow \infty$ $\sqrt{n} (\hat{\theta} - \theta_T | X_1, \dots, X_n) \rightarrow N(0, \Sigma)$

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Write the code in rstan package.

generates $X_1, \dots, X_n \stackrel{iid}{\sim}$ Double Exponential(θ)

~~$X_1, \dots, X_n \stackrel{iid}{\sim}$~~

b) Same as (a) only now $\sqrt{n} (\hat{\theta}^2 - \theta_T^2 | X_1, \dots, X_n) \rightarrow N(0, \Sigma)$

Q3 (3 marks)

$X_1, \dots, X_n \stackrel{iid}{\sim} pN(\mu_1, 1) + (1-p)N(\mu_2, 1)$

True values $(p = 0.2, \mu_1 = 0, \mu_2 = 4)$

Then using simulation find the ~~dis~~ posterior distⁿ of p, μ_1, μ_2

$\pi(\mu_1) = \pi(\mu_2)$

$0.2N(0, 1) + 0.8N(4, 1)$

~~$0.8N(4, 1) + 0.2N(0, 1)$~~ ~~$p = 0.8$~~ ~~$\mu_1 = 4$~~ ~~$\mu_2 = 0$~~

(1) Markov Chain

(2) Gibbs sampling

(3) Metropolis - Hastings

Diagnostics of convergence

Linear Model
Hierarchical

Data analysis

$f(x, y, z)$

$f(x, y, z)$

$\pi(\theta) p(\gamma | \theta)$

